Problem-Solving & Decision-Making

Updated FY2026

Use the following slide to break the ice and to introduce the concept of critical thinking, a skill essential to problem solving and decision making.

Critical Thinking

Ask the following questions. Let the students try to answer as you go.

Q: How do you put a giraffe into the refrigerator?

A: Open the refrigerator, put in the giraffe, and close the door.

(This question tests whether you tend to do simple things in an overly complicated way.)

Q: How do you put an elephant into the refrigerator?

("Open the refrigerator, put in the elephant, and close the refrigerator" is not the correct answer.)

A: Open the refrigerator, take out the giraffe, put in the elephant, and close the door.

(This question tests your ability to think through the repercussions of your previous actions.)

Q: The Lion King is hosting an animal conference. All the animals are in attendance, except one---which one?

A: The elephant---who is in the refrigerator. You just put him in there.

(This question tests your memory.) Okay, even if you did not answer the first three questions correctly, you still have one more chance to show your stuff.

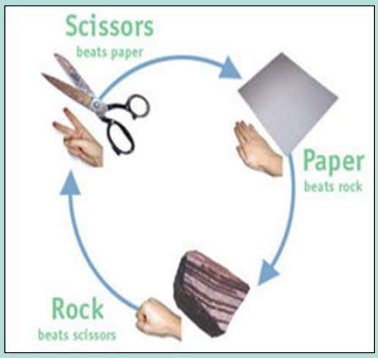
Q: You must cross a river used by crocodiles, and you do not have a boat. How do you do it?

A: You jump into the river and swim across. All the crocodiles are attending the animal conference, remember?

(This tests whether you learn quickly from your mistakes.)

Per Anderson Consulting Worldwide, around 90% professionals tested got all wrong. The group that got the most correct were children!

Objectives



- To participate in discussions about problem-solving and decision-making
  Participate in discussions about problem solving and decision making
- To recognize the correlation between critical thinking and problem-solving
  *To recognize the correlation between critical thinking and problem solving
- To identify personal decision-making style(s)
  *To identify their personal decision making styles
- To learn to apply a logical method in solving problems and making decisions
  *To learn to apply a logical method in solving problems and making decisions

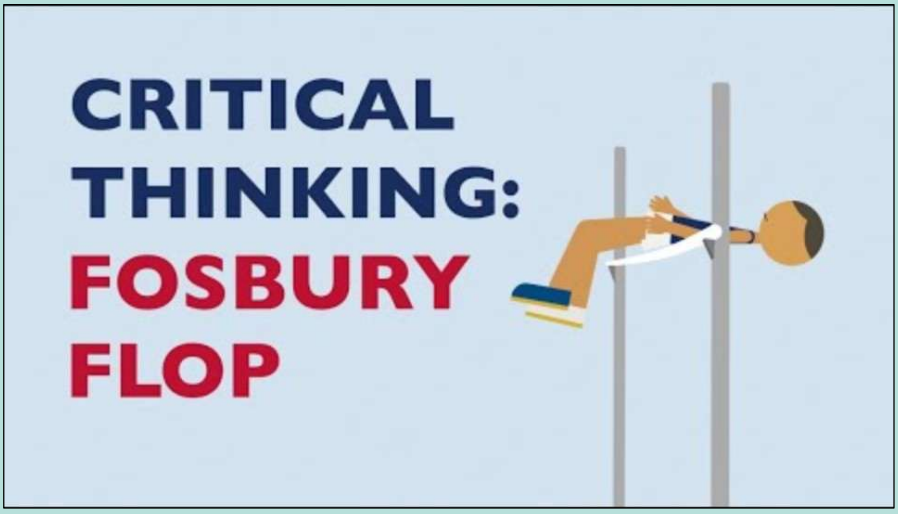
Think about a small decision you have made recently. Then think about the last major decision you had to make. Which one was easier to recall? Which one had the most impact on your life? Our days are filled with all kinds of decisions, some more impactful than others.

Today we are going to explore approaches and processes to help us make decisions in an effective way. And what usually lies behind the decisions we have to make? A particular set of circumstances---in many cases, a problem to solve.

*Does this decision impact my family in the short-term or long-term?
*How might this decision affect our lives?

Not all decisions are hard to make, and they will not impact the future drastically (example: shirt color). However, problems will arise that require us to make decisions that will impact us now and in the future. If we approach problems using critical thinking methods, we can use our knowledge and experience to make effective decisions.

Fosbury Flop



Start a discussion on how Dick Fosbury solve a problem by thinking critically. Ask students how the results of his decision affected himself and others.

Problem-Solving Styles

Introduce and distribute the Problem-Solving Styles Questionaire, and review the directions. Ask students to use their experience in problem-solving from the previous activity and apply it to this one. Complete the first question aloud to model the ranking process, and allow learners to se how to properly score and document their responses. Allow students time to complete the activity. Discuss results and whether or not students agree or disagree with their styles. Share the next slides for explanation of the styles.

Professor

* Interested in theories

* Seeks lots of background information

* Learns from relationships to other situations

* Values objectivity and detachment from issues

* Relates to others by giving and getting information

** Needs to guard against indecision and impracticality (always wanting more information)

Diplomat

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* Interested in reaching consensus (agreement)
* Learns best from talking to those involved
* Often values opinions more than facts
* Focuses on practical solutions
* Can be very persuasive

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Detective

* Interested in finding the root cause of the problem

* Focuses on relevant facts

* Prefers to make direct observations of situations

* Tends to follow a consistent method or process

* Tries to establish clear cause-and-effect relationships

** Needs to guard against inflexibility and insensitivity to people

Champion

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* Interested in results

* Finds a solution by trying one out

* Wants to achieve success without help from others

* Trusts his or her instincts

* Good in crisis situations when there is no time for analysis

** Needs to guard against impulsiveness and uncooperativeness**

Four Step Problem-Solving/Decision-Making Process

Step 1

Stop!

Think about the problem

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When presented with a problem to solve or a decision to make, we sometimes react in an extreme manner – either by being impulsive or by procrastinating. What might happen if we react impulsively and make decisions very quickly without much thought and considerations? What might happen if we procrastinate in making a decision? How could the outcomes be different if we stopped first, then thought about it, then made the decision?

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Step 2

What are some options?

One great technique in exploring problem-solving and decision-making is to write a list of possible solutions…. Also called options or choices. We create our list of options by brainstorming random solutions, even if they are “outside the box.” After our list is full, we then eliminate those choices that won’t solve the problem in an ethical, legal, or positive way. For example, if our options include breaking the law or harming someone’s property, we would mark it out as an unrealistic solution.

Step 3

Choose one

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As we look at the list in progress, we may have only one or two possible choices left. In the case when there are two choices from which to pick, select the one that is most effective without causing negative effects. If there is only one viable suggestion left, it may be the best option.

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Step 4

Do it.

Apply the remaining option to your situation.

Four Step Process "Scenario"

Example:

Your kids need new shirts for school, and with a limited budget, you look for a bargain on your favorite store's website.

You find a great sale on children's shirts at 3 for $10.00!

So, you rush down to the store to buy them before they're gone. You find the same shirts listed at $15.00 each – no sale sign in sight!

What could you do???

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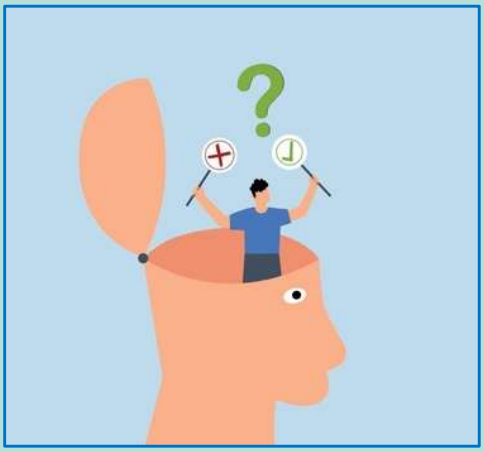
Present this scenario and ask students to stop and think about the problem/decision. Ask students for option suggestions to create a list on the board. You may need to make a couple of suggestions "outside the box" to illustrate the procedure. After creating a list with several possible solutions, ask students to start eliminating (cross them out) suggestions that won't work and give reasons why. Announce the only remaining option, this is the solution to the problem or decision made.

Practice 1 Scenario

- Divide into Groups
- Choose a Scenario
- Discuss
- Make a List
- Eliminate Suggestions
- Share Results



Divide the class into 2 or more groups. Students select a scenario (*see module handouts*) and choose a scribe per each group. Group members stop and think about the problem/decision. Discuss. Scribe asks for option suggestions to create a list on paper. After creating a list with several possible solutions, ask students to start eliminating suggestions that won't work (cross out these choices) and give reasons. When groups have finished, share the results with the class.

Practice 2
Survival Activity

- Choose a Survival Activity
- Rank Items by Necessity
- Reveal the Appropriate Answers
- Discuss Reasons for Rankings

Instructor asks students to select one of the several survival activities (*see handouts*) – **Lost at Sea**, **Lunar Survival,** or **Stranded on a Mountain** to practice collaborative problem-solving and decision-making. Explain the ranking method process. On the board list the items necessary to rank (without explanations). Ask students to brainstorm, use the four-step process to determine their answers, and write rankings next to items. When they've completed the activity, reveal the appropriate answers from the activity answer key. Discuss the reasons for the rankings.

Evaluation/Application

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3 – 2 – 1 Reflection

3 – Things you learned about problem-solving & decision-making

2 – Barriers you want to improve on

1 – Strategy you can implement today

Teacher observation of learners' comprehension of concepts illustrated in activities. Complete the rubric for each learner.

Learners complete and share their responses to the 3-2-1 Reflection.